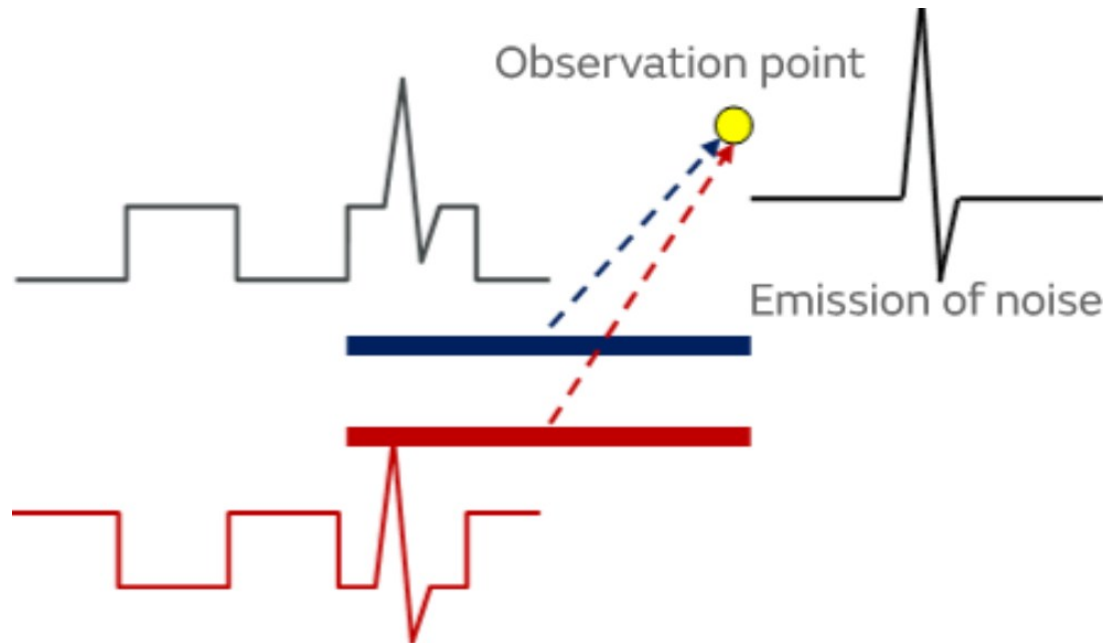
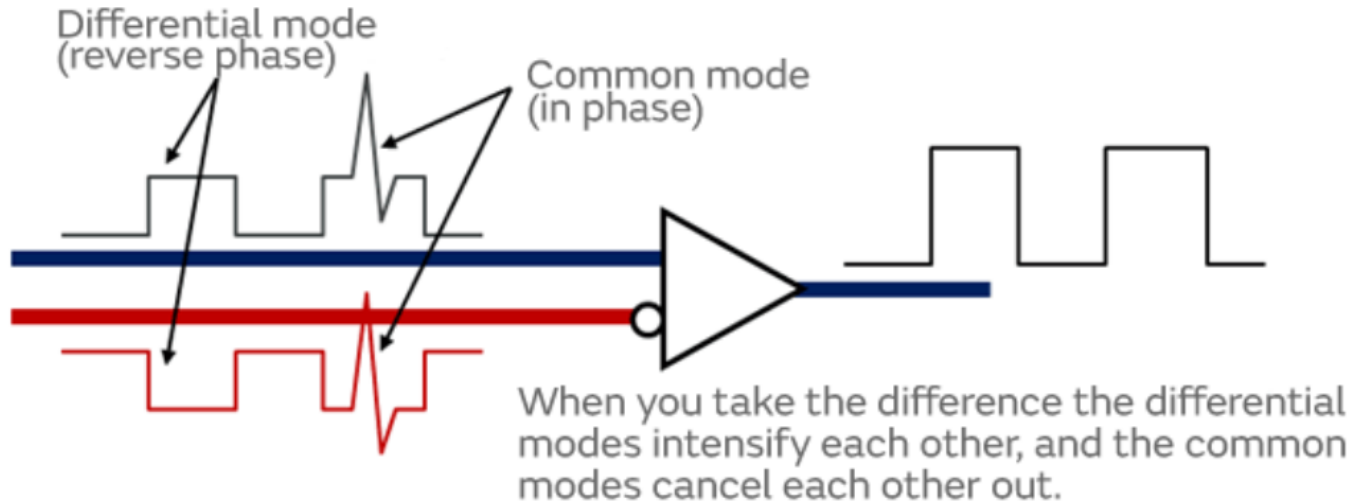


Differential Amplifiers

Why differential?

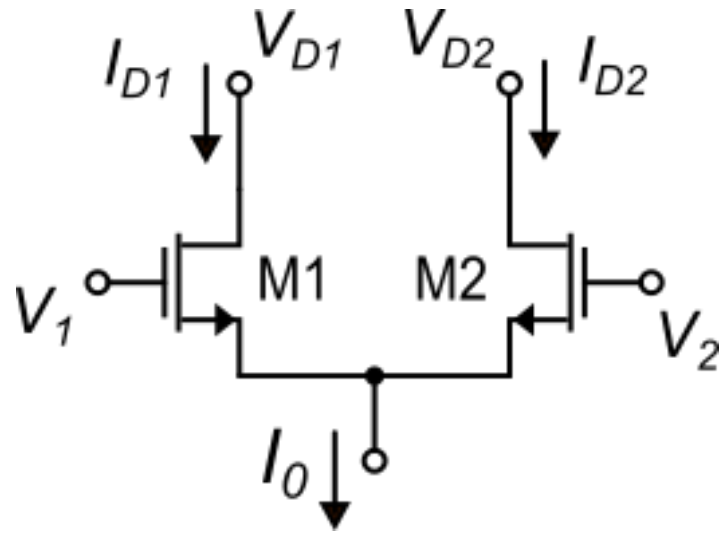
Example Applications?

Differential Signals



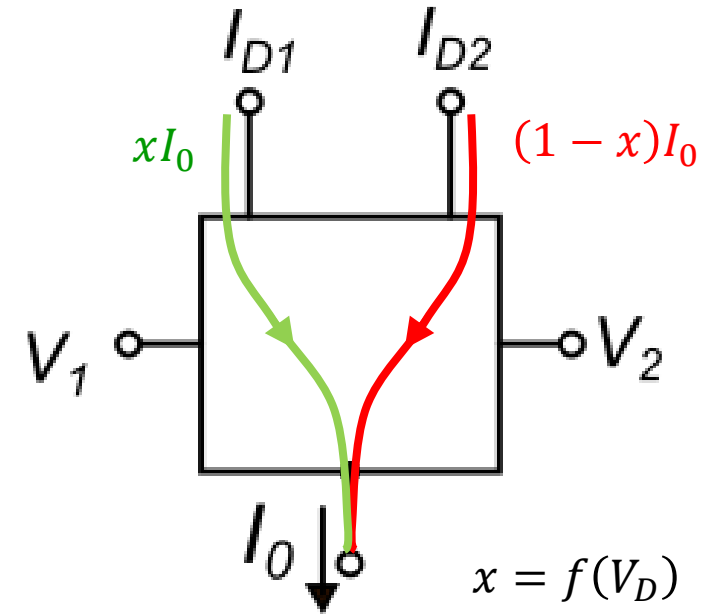
This signal, called the differential mode signal, is used for the transmission of data (the differential mode is sometimes called the normal mode). Meanwhile, there is also a so-called "common mode" signal, which refers to the signal flowing in phase through the two lines. The common mode signal is unwanted from the perspective of differential transmission. In other words, it is a form of noise, called "common-mode noise".

Source coupled MOSFET pair (differential MOSFET pair)



Inputs: V_1, V_2 (effective input signal: $V_D = V_1 - V_2$)
 I_0 ("Tail current")

Outputs: I_{D1}, I_{D2} ($I_{D1} - I_{D2}$)



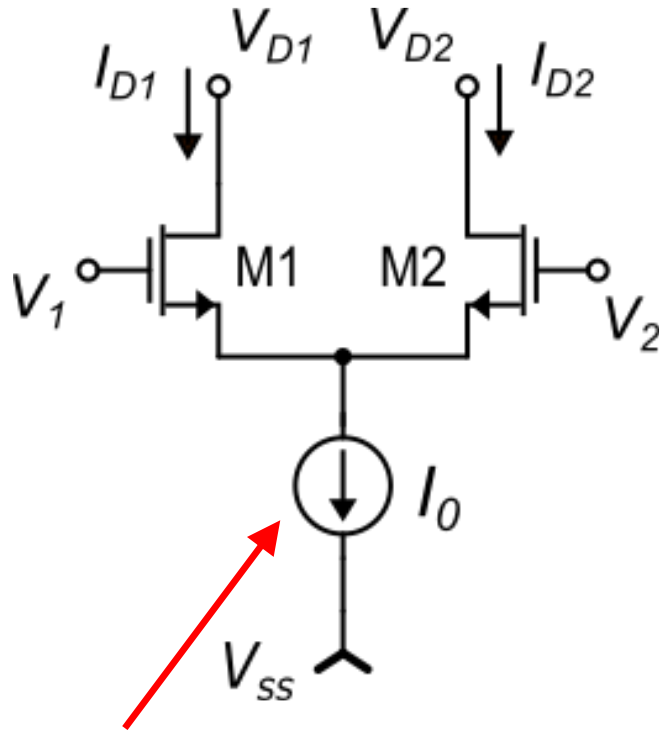
Function

- The input current I_0 is split into I_{D1} and I_{D2} , according to fractions x and $(1-x)$
- x depends on V_D ($x=0.5$ for $V_D=0$)

Analysis of the source coupled MOSFET pair (differential MOSFET pair)

Target: obtain the relationship between the ratio x and the input differential voltage V_D

$$I_{D1} = xI_0 \quad I_{D2} = (1 - x)I_0 \quad \Rightarrow \quad x \triangleq \frac{I_{D1}}{I_0}$$



Hypotheses

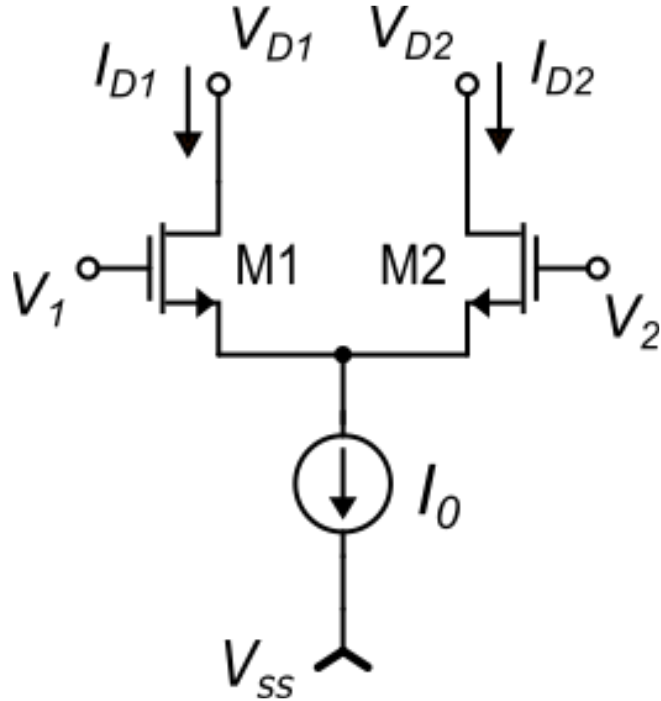
- M1 and M2 work in saturation and the effect of V_{DS} can be neglected
- Strong inversion equation can be applied
- I_0 does not depend on V_1, V_2

I_0 has been represented as an ideal current source placed between M1, M2 sources and V_{SS} : this is just an example.

$$I_D = \frac{\beta}{2} (V_{GS} - V_t)^2 \Rightarrow V_{GS} = V_t + \sqrt{\frac{2I_D}{\beta}}$$

$$\beta = \frac{W\mu_n C_{ox}}{L}$$

Analysis of the source coupled MOSFET pair (differential MOSFET pair)



$$\begin{cases} V_1 = V_{S1} + V_{GS1} \\ V_2 = V_{S2} + V_{GS2} \end{cases} \quad V_{S1} = V_{S2} \triangleq V_S$$

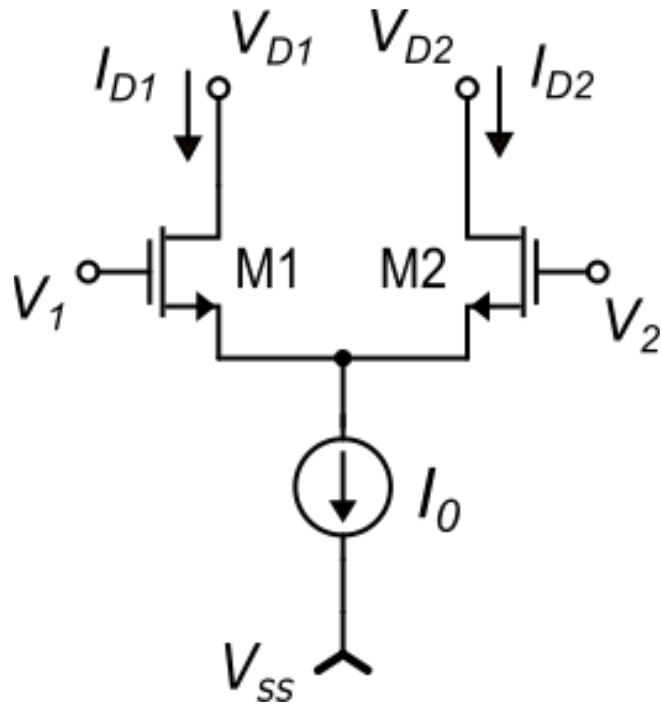
$$V_D = V_1 - V_2 = V_{GS1} - V_{GS2}$$

$$V_D = V_{t1} + \sqrt{\frac{2I_{D1}}{\beta_1}} - \left(V_{t2} + \sqrt{\frac{2I_{D2}}{\beta_2}} \right)$$

Nominal conditions: $M1=M2$: $\beta_1=\beta_2=\beta$ and $V_{t1}=V_{t2}$

$$V_D = \sqrt{\frac{2I_{D1}}{\beta}} - \sqrt{\frac{2I_{D2}}{\beta}} = \sqrt{\frac{2}{\beta}} \cdot (\sqrt{I_{D1}} - \sqrt{I_{D2}})$$

Analysis of the source coupled MOSFET pair (differential MOSFET pair)



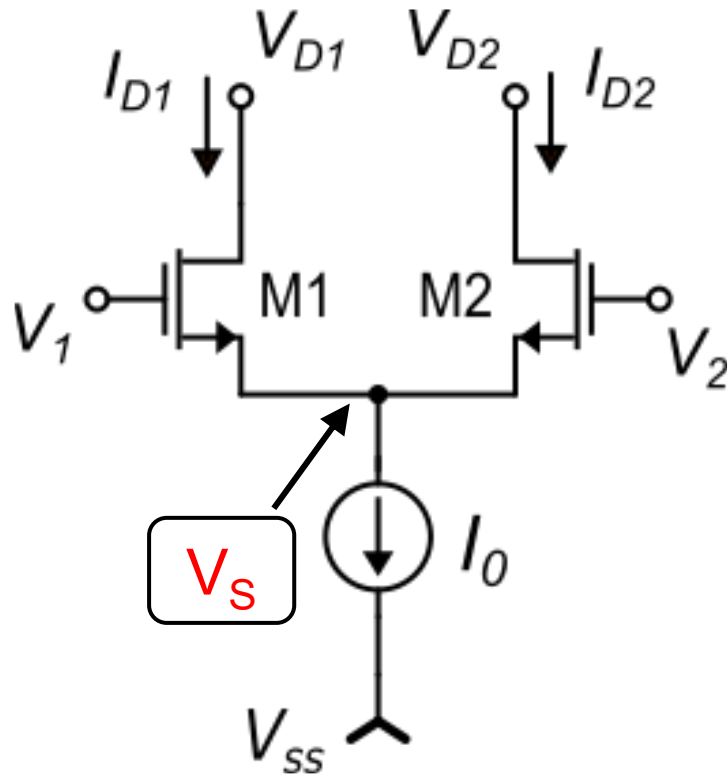
$$V_D = \sqrt{\frac{2I_{D1}}{\beta}} - \sqrt{\frac{2I_{D2}}{\beta}} = \sqrt{\frac{2}{\beta}} \cdot (\sqrt{I_{D1}} - \sqrt{I_{D2}})$$

$$V_D = \sqrt{\frac{2}{\beta}} \cdot [\sqrt{xI_0} - \sqrt{(1-x)I_0}] = \boxed{\sqrt{\frac{2I_0}{\beta}}} \cdot [\sqrt{x} - \sqrt{(1-x)}]$$

$$\frac{V_D}{V_{D\text{MAX}}} = [\sqrt{x} - \sqrt{(1-x)}]$$

$$\sqrt{\frac{2I_0}{\beta}} \triangleq V_{D\text{MAX}}$$

Large-signal dependence of the source voltage on V_D



$$V_1 = V_C + \frac{V_D}{2}; V_2 = V_C - \frac{V_D}{2}$$

V_C : common mode voltage=*constant*

$$V_S = V_1 - V_{GS1} = V_2 - V_{GS2}$$

V_S is a virtual ground since the AC voltage at this node is equal to zero

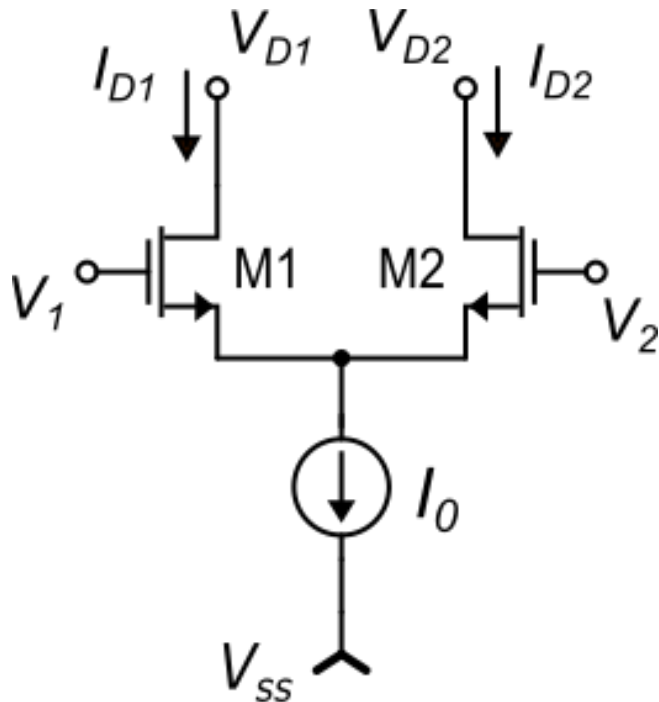
$$I_{D1} = \frac{I_0}{2} + \frac{g_m}{2} v_d$$

$$I_{D2} = \frac{I_0}{2} - \frac{g_m}{2} v_d$$

$$I_{D1} - I_{D2} = g_m v_d$$

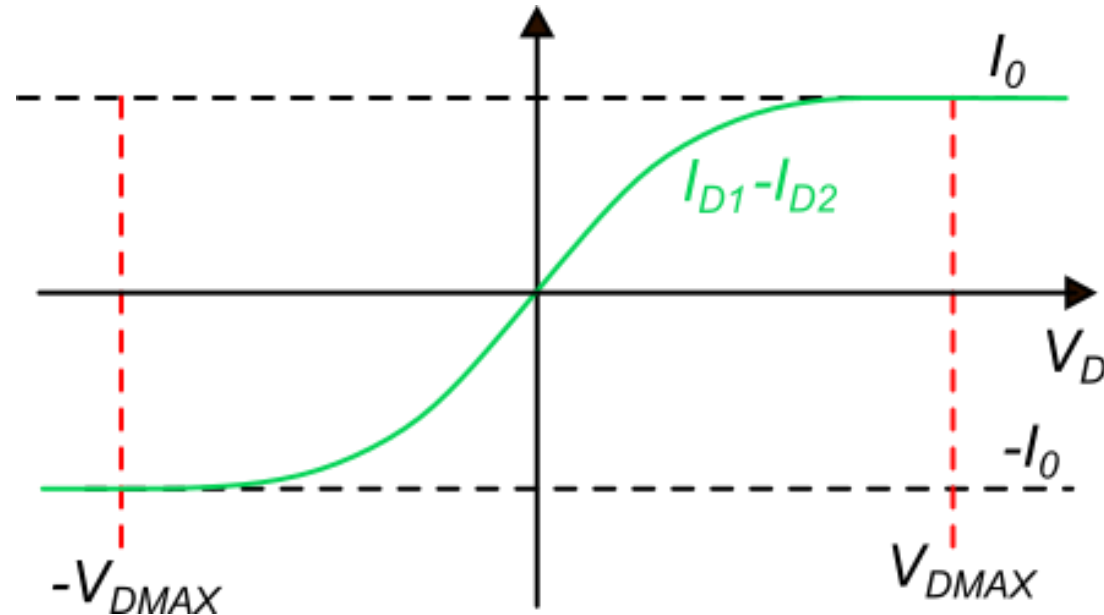
$$I_{D1} - I_{D2} = I_0 \left(\frac{V_D}{V_{DMAX}} \right) \sqrt{2 - \left(\frac{V_D}{V_{DMAX}} \right)^2}$$

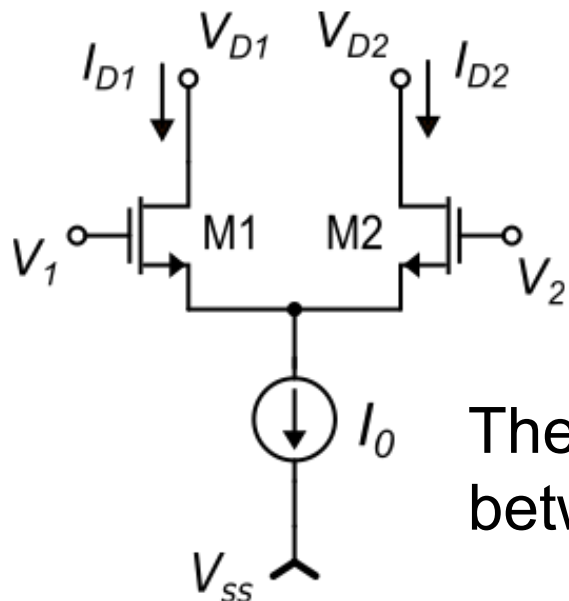
$$i_d = i_{d1} - i_{d2} = g_m v_d$$



$$I_{D1} = \frac{I_0}{2} + \frac{I_0}{2} \left(\frac{V_D}{V_{DMAX}} \right) \sqrt{2 - \left(\frac{V_D}{V_{DMAX}} \right)^2}$$

$$I_{D2} = \frac{I_0}{2} - \frac{I_0}{2} \left(\frac{V_D}{V_{DMAX}} \right) \sqrt{2 - \left(\frac{V_D}{V_{DMAX}} \right)^2}$$





$$\left(\frac{V_D}{V_{D\text{MAX}}} \right)^2 - 1 \leq 0$$

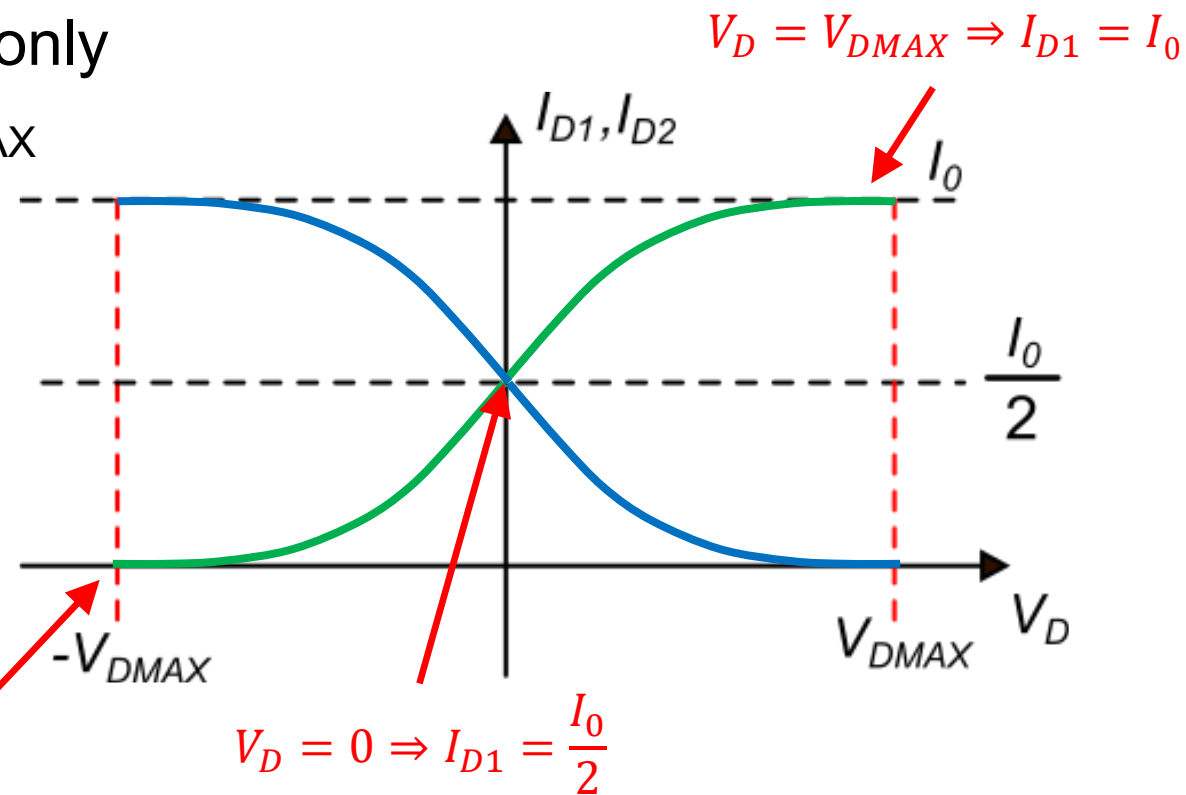


$$\left| \frac{V_D}{V_{D\text{MAX}}} \right| \leq 1 \Rightarrow -V_{D\text{MAX}} \leq V_D \leq V_{D\text{MAX}}$$

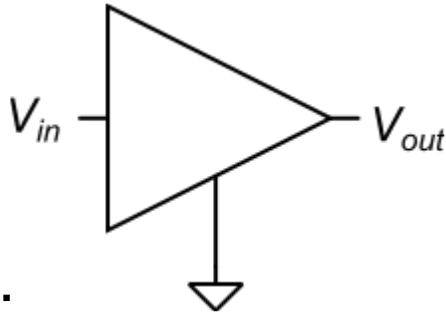
The analysis is applicable only between $-V_{D\text{MAX}}$ and $+V_{D\text{MAX}}$

$$I_{D1} = \frac{I_0}{2} + \frac{I_0}{2} \left(\frac{V_D}{V_{D\text{MAX}}} \right) \sqrt{2 - \left(\frac{V_D}{V_{D\text{MAX}}} \right)^2}$$

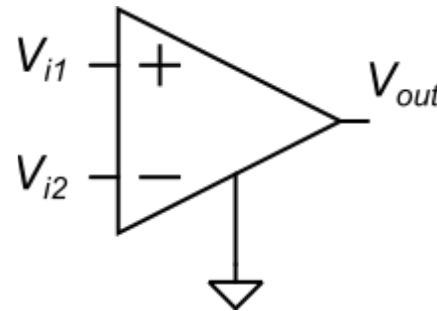
$$I_{D2} = \frac{I_0}{2} - \frac{I_0}{2} \left(\frac{V_D}{V_{D\text{MAX}}} \right) \sqrt{2 - \left(\frac{V_D}{V_{D\text{MAX}}} \right)^2}$$



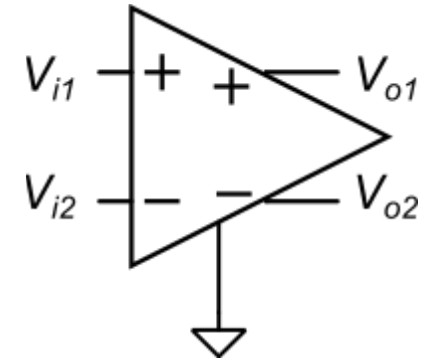
Voltage amplifiers: number of inputs and outputs



Unipolar:
single input
single output (single-ended, S/E)



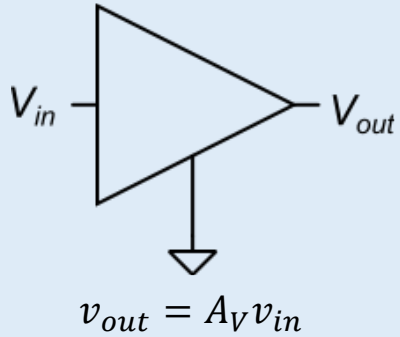
Differential:
differential input,
single output (single-ended, S/E)



Fully Differential:
differential input,
differential output

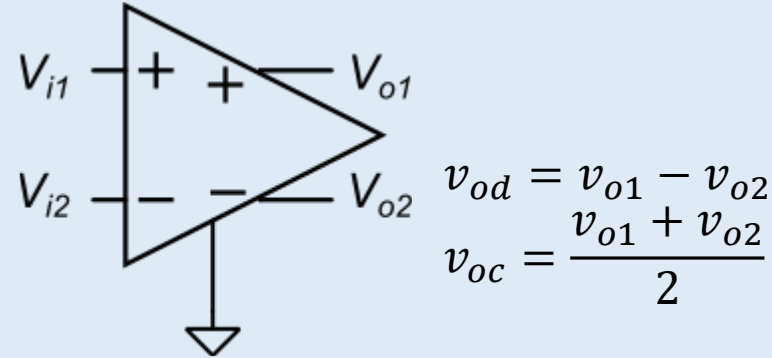
Voltage Amplifiers: gains

Differential Output:



$$v_{id} = v_{i1} - v_{i2}$$

$$v_{ic} = \frac{v_{i1} + v_{i2}}{2}$$



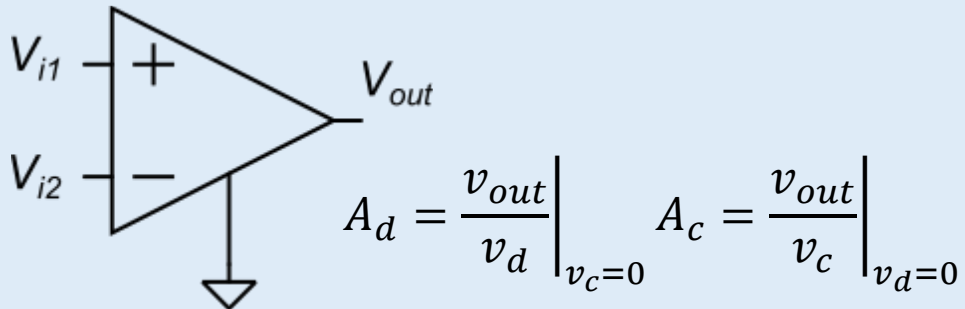
$$A_{dd} = \left. \frac{v_{od}}{v_{id}} \right|_{v_{ic}=0}$$

$$A_{cd} = \left. \frac{v_{od}}{v_{ic}} \right|_{v_{id}=0}$$

$$A_{dc} = \left. \frac{v_{oc}}{v_{id}} \right|_{v_{ic}=0}$$

$$A_{cc} = \left. \frac{v_{oc}}{v_{ic}} \right|_{v_{id}=0}$$

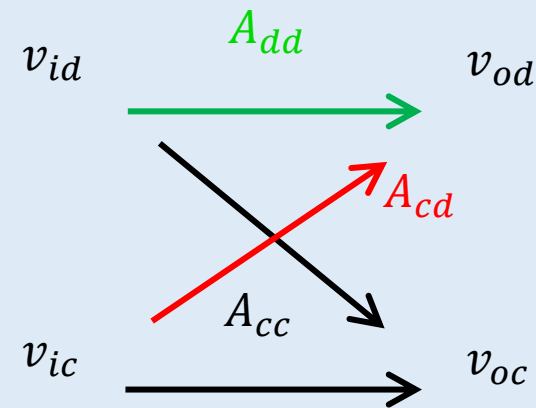
Single-ended Output:



$$v_d = v_{i1} - v_{i2}$$

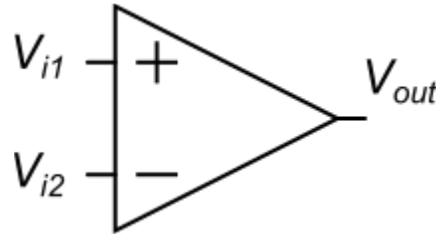
$$v_c = \frac{v_{i1} + v_{i2}}{2}$$

$$CMRR = \left| \frac{A_d}{A_c} \right|$$



$$CMRR = \left| \frac{A_{dd}}{A_{cd}} \right|$$

Differential amplifiers: parameters



$$V_{out} = A_d(V_{id} - V_{io})$$

Linear response
(with input offset voltage)

Input voltage ranges:

Input differential range
(maximum V_{ID} (V_D) to
maintain an acceptable
input-output linearity

$$-V_{DMAX} \leq V_D \leq V_{DMAX}$$

Input common-mode range:

Interval of V_C values where
the amplifier behaves as
designed

$$V_{CMIN} \leq V_C \leq V_{CMAX}$$

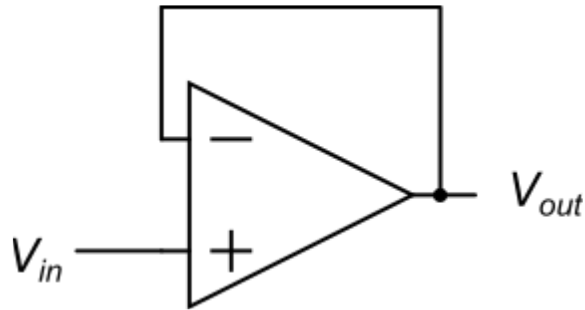
Output voltage range (Output voltage swing)

$$V_{OMIN} \leq V_{out} \leq V_{OMAX}$$

Outside the output range
the amplifier stop working
correctly (e.g. the output
voltage tends to saturate)

Differential amplifiers: role of the input common mode range

Example: voltage follower (buffer amplifier)



In the ideal case:

$V_{io} = 0, A_d \rightarrow \infty$ (V_{io} : Input Offset Voltage due to tolerances, mismatch etc.)

We have:

$$V_C = V_{in}$$

$$V_D \cong 0$$



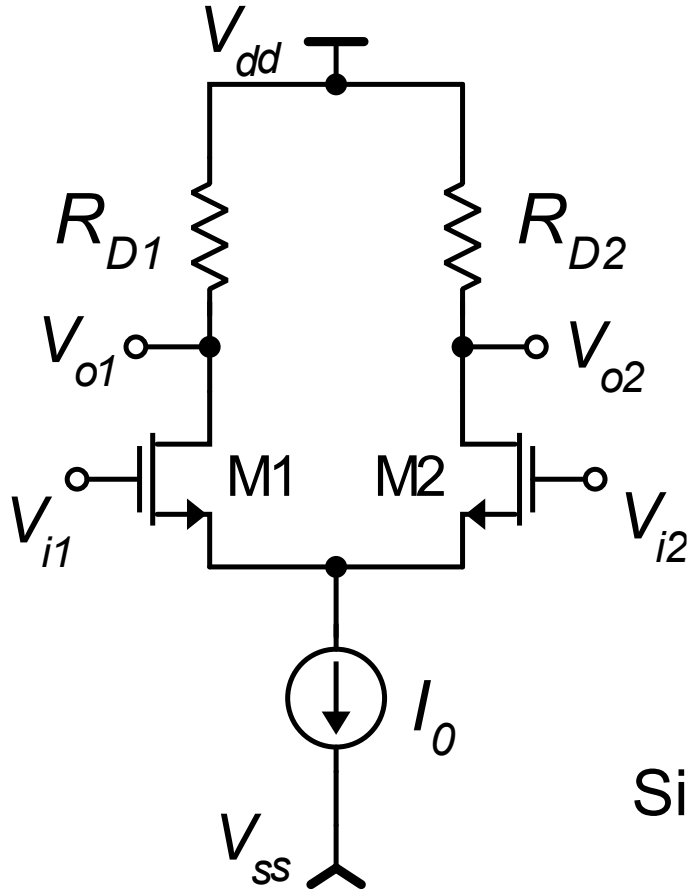
The voltage follower operates correctly only for V_{in} within the CM range

Another condition is clearly that V_{out} is within the output range

Then the conditions for correct operation are:

$$V_{out} = V_{in} \text{ only for } \begin{cases} V_{CMIN} \leq V_{in} \leq V_{CMAX} \\ V_{OMIN} \leq V_{out} \leq V_{OMAX} \end{cases}$$

Differential amplifier with resistive loads



$$V_{ID} = V_{i1} - V_{i2} \quad V_{OD} = V_{o1} - V_{o2}$$

$$V_{O1} = V_{DD} - R_{D1}I_{D1} \quad V_{O2} = V_{DD} - R_{D2}I_{D2} \quad V_{OD} = R_{D2}I_{D2} - R_{D1}I_{D1}$$

Small signal analysis

$$v_{o1} = -R_{D1}i_{d1} \quad v_{o2} = -R_{D2}i_{d2} \quad v_{od} = i_{d2}R_{D2} - i_{d1}R_{D1}$$

$$\text{Differential mode:} \quad i_{d1} = \frac{g_m}{2}v_{id} \quad i_{d2} = -\frac{g_m}{2}v_{id}$$

Single-ended case:

$$v_{out} = v_{o1} = -R_{D1}\frac{g_m}{2}v_{id}$$

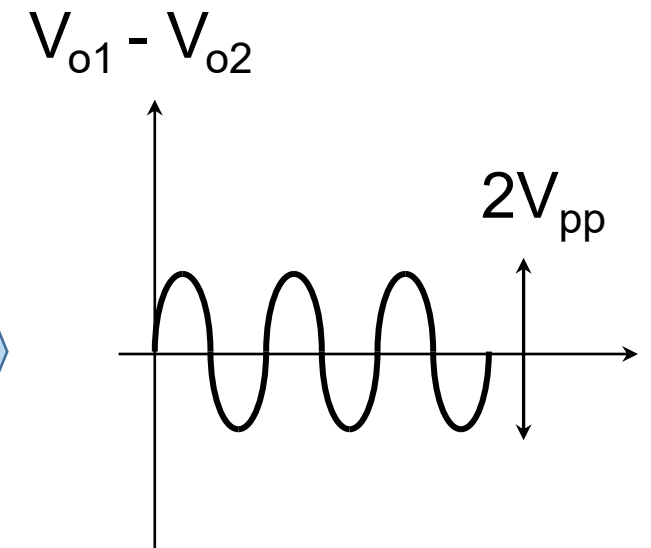
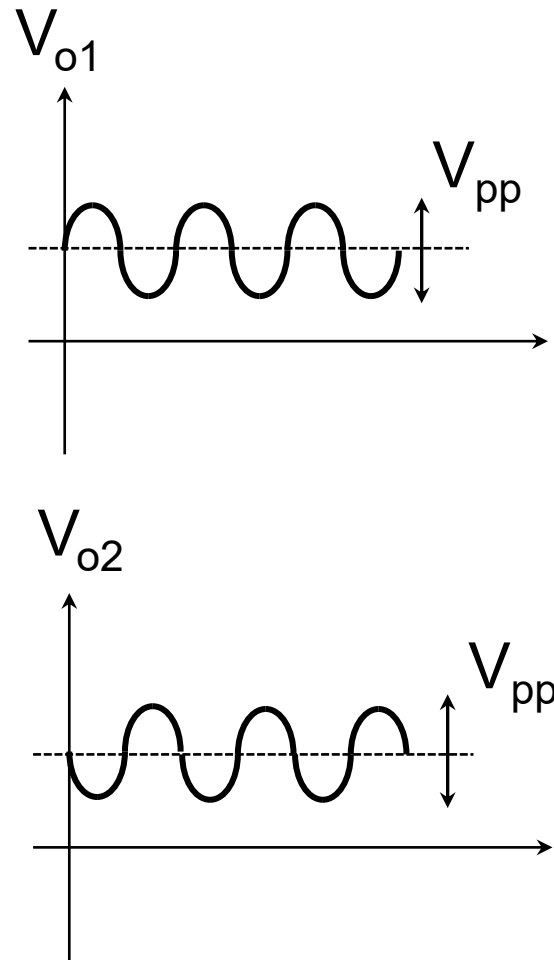
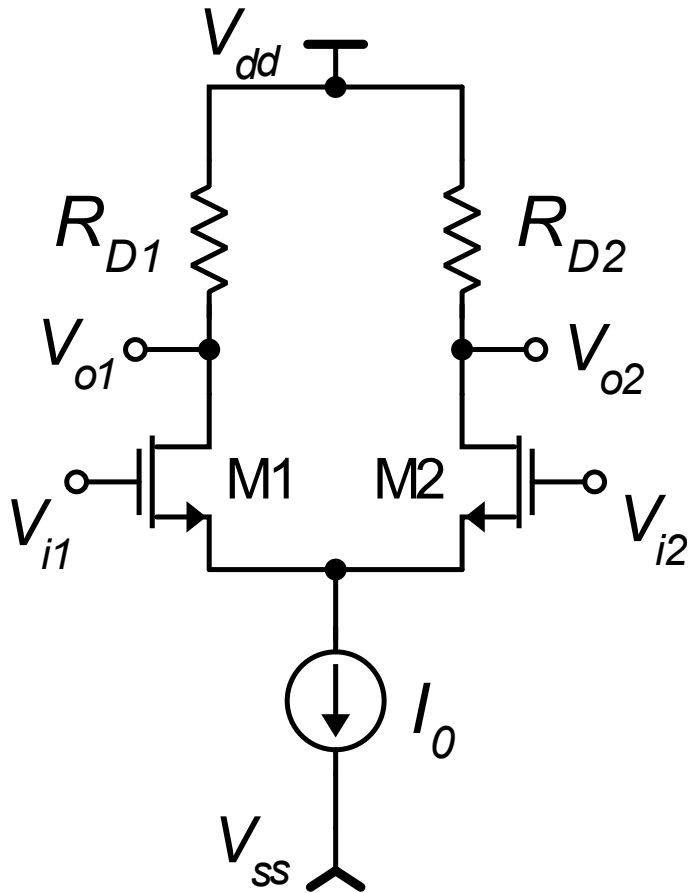
$$A_d = -R_{D1}\frac{g_m}{2}$$

Fully-differential case:

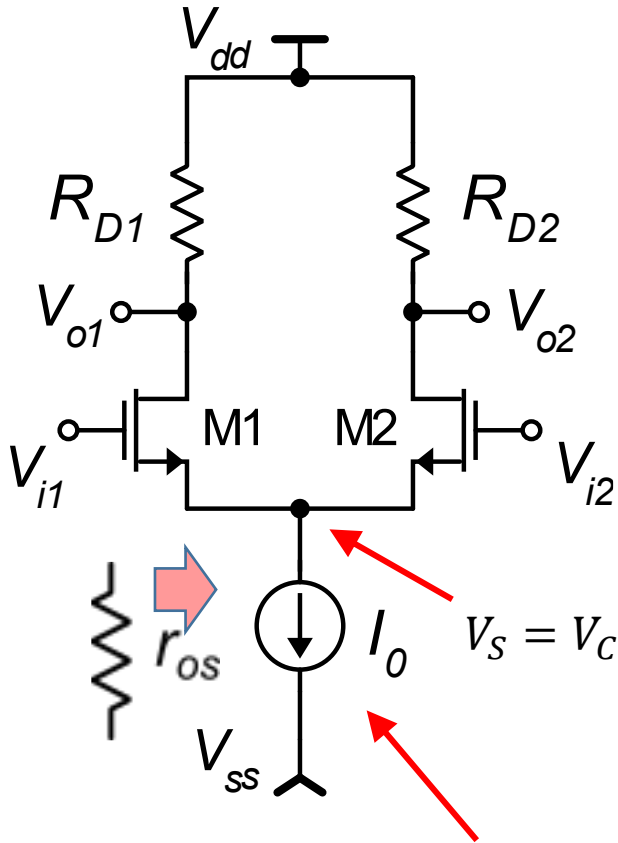
$$v_{out} = v_{od} = -\frac{g_m}{2}(R_{D1} + R_{D2})v_{id}$$

$$A_{dd} = -g_m\frac{(R_{D1} + R_{D2})}{2}$$

Differential amplifier with resistive loads



Common mode Analysis



$$v_{id} = 0 \Rightarrow i_{d1} = i_{d2}$$

$$v_{i1} = v_{i2} = v_c$$

$$i_0 = i_{d1} + i_{d2} = \frac{v_s}{r_{os}} \quad * \text{ These are small-signal voltages, currents and resistances!}$$

$$v_s = v_{i1} - v_{gs1} = v_c - v_{gs1}$$

$$i_{d1} \cong g_m v_{gs1}$$

$$V_{GS} \cong \text{constant}$$

$$v_{gs1} = \frac{i_{d1}}{g_m}$$

$$v_s = v_c - \frac{i_{d1}}{g_m}$$

$$i_{d1} + i_{d2} = 2i_{d1} = i_0$$

$$i_{d1} = \frac{i_0}{2} = \frac{v_s}{2r_{os}}$$

$$v_s = v_c - \frac{v_s}{2r_{os}g_m}$$

This, actually, is not an ideal current source. It has a finite output resistance = r_{os} . If it is implemented by a simple current mirror, $r_{os} = r_d$ otherwise it can be even higher

The source voltage practically follows the common mode variations. V_S is V_C shifted by V_{GS}

Common mode gain and CMRR

output voltages

$$i_{d1} = i_{d2} \cong \frac{i_o}{2} \quad v_{o1} = -R_{D1}i_{d1} \quad v_{o2} = -R_{D2}i_{d2} \quad v_{od} = i_{d2}R_{D2} - i_{d1}R_{D1}$$

Single-ended case

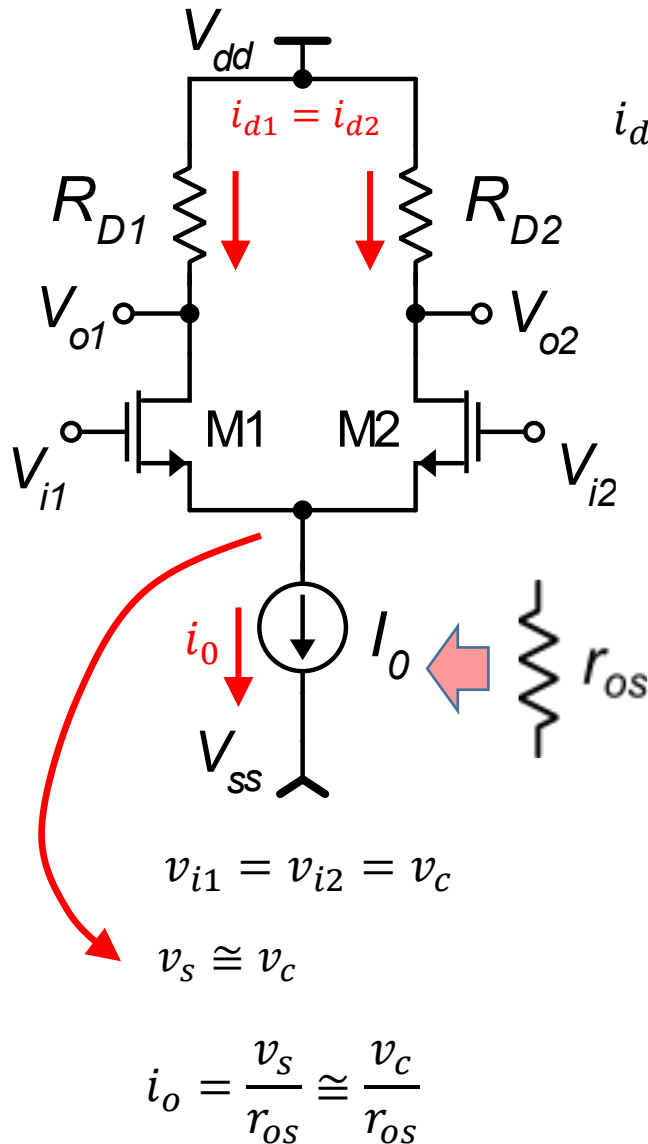
$$v_{out} = v_{o1} = -R_{D1}i_{d1} \quad v_{out} \cong -R_{D1} \frac{v_c}{2r_{os}}$$

$$A_c = \left. \frac{v_{out}}{v_c} \right|_{v_d=0} \cong -\frac{R_{D1}}{2r_{os}}$$

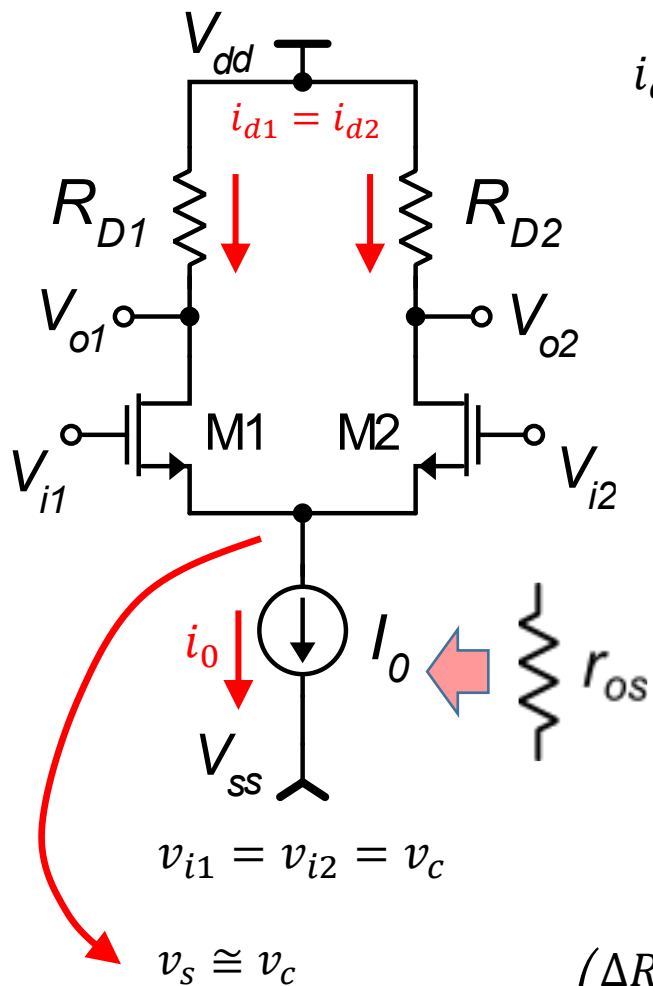
$$CMRR = \frac{A_d}{A_c} = \frac{g_m R_{D1}}{2} \frac{2r_{os}}{R_{D1}} \quad |A_d|$$

Considering that generally I_o is produced by a simple current mirror, $r_{os}=r_d$, then $CMRR \sim 40$ dB,

This CMRR is not sufficient for many applications



Common mode gain and CMRR



$$i_{d1} = i_{d2} \cong \frac{i_0}{2}$$

$$v_{o1} = -R_{D1}i_{d1}$$

$$v_{o2} = -R_{D2}i_{d2}$$

$$v_{od} = i_{d2}R_{D2} - i_{d1}R_{D1}$$

Fully differential case

$$v_{out} = v_{od} = \frac{i_0}{2}R_{D2} - \frac{i_0}{2}R_{D1} = \frac{i_0}{2}(R_{D2} - R_{D1})$$

$$v_{out} = \frac{v_c}{2r_{os}}(R_{D2} - R_{D1})$$

$$A_{cd} = \left. \frac{v_{od}}{v_{ic}} \right|_{v_{id}=0} = -\frac{1}{2r_{os}}(R_{D1} - R_{D2})$$

$$CMRR = \left| \frac{A_{dd}}{A_{cd}} \right| = \frac{\frac{g_m}{2}(\bar{R}_D)}{\frac{2r_{os}}{\Delta R_D}} = 2g_m r_{os} \left(\frac{\Delta R_D}{\bar{R}_D} \right)^{-1}$$

$\bar{R}_D$ ΔR_D

$\left(\frac{\Delta R_D}{\bar{R}_D} \right)$ is a relative matching error ~ 0.01

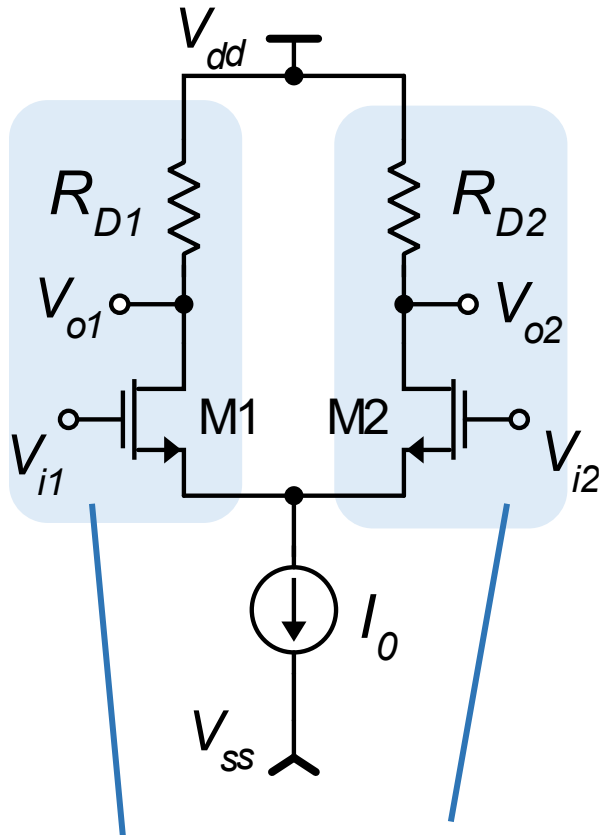
$\Rightarrow \left(\frac{\Delta R_D}{\bar{R}_D} \right)^{-1} \approx 100$

$$CMRR \approx 10^4 = 80\text{dB}$$

Adequate for most applications

Offset voltage of differential amplifiers

MOSFET case - fully-differential - Strong inversion



Nominally identical
circuits

$$v_{io} = V_D \Big|_{V_{out}=0}$$

$$V_{out} = V_{o1} - V_{o2}$$

$$V_{GS} = V_t + \sqrt{\frac{2I_D}{\beta}}$$

$$V_{GS} = A + B$$

$$\Delta A = \Delta V_t$$

$$V_D = V_{GS1} - V_{GS2} \triangleq \Delta V_{GS}$$

We can consider the difference of V_{GS} as the matching error between two nominally identical circuits: M1, R_{D1} and M2, R_{D2}

$$A = V_t \quad B = \sqrt{\frac{2I_D}{\beta}}$$

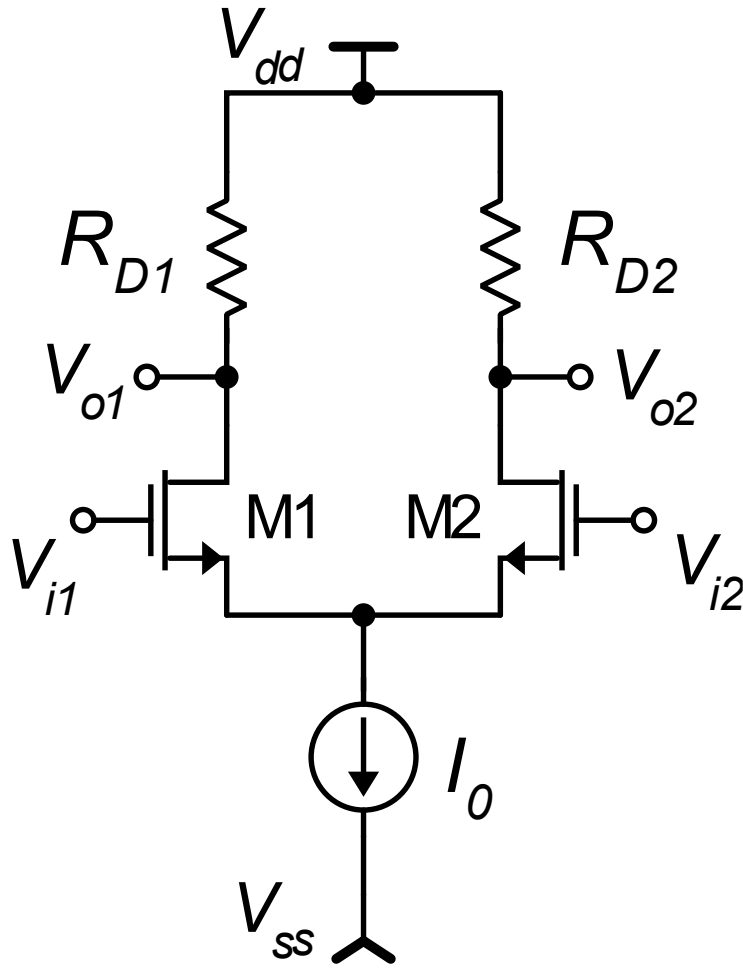
$$\Delta V_{GS} = \Delta A + \Delta B$$

Differences in W , L , C_{ox} or μ_n between M1 and M2 lead to a larger ΔB .

This expression gives V_{io} for the particular ΔV_{GS} value that results in $V_{out}=0$

Offset voltage of differential amplifiers

MOSFET case - fully-differential

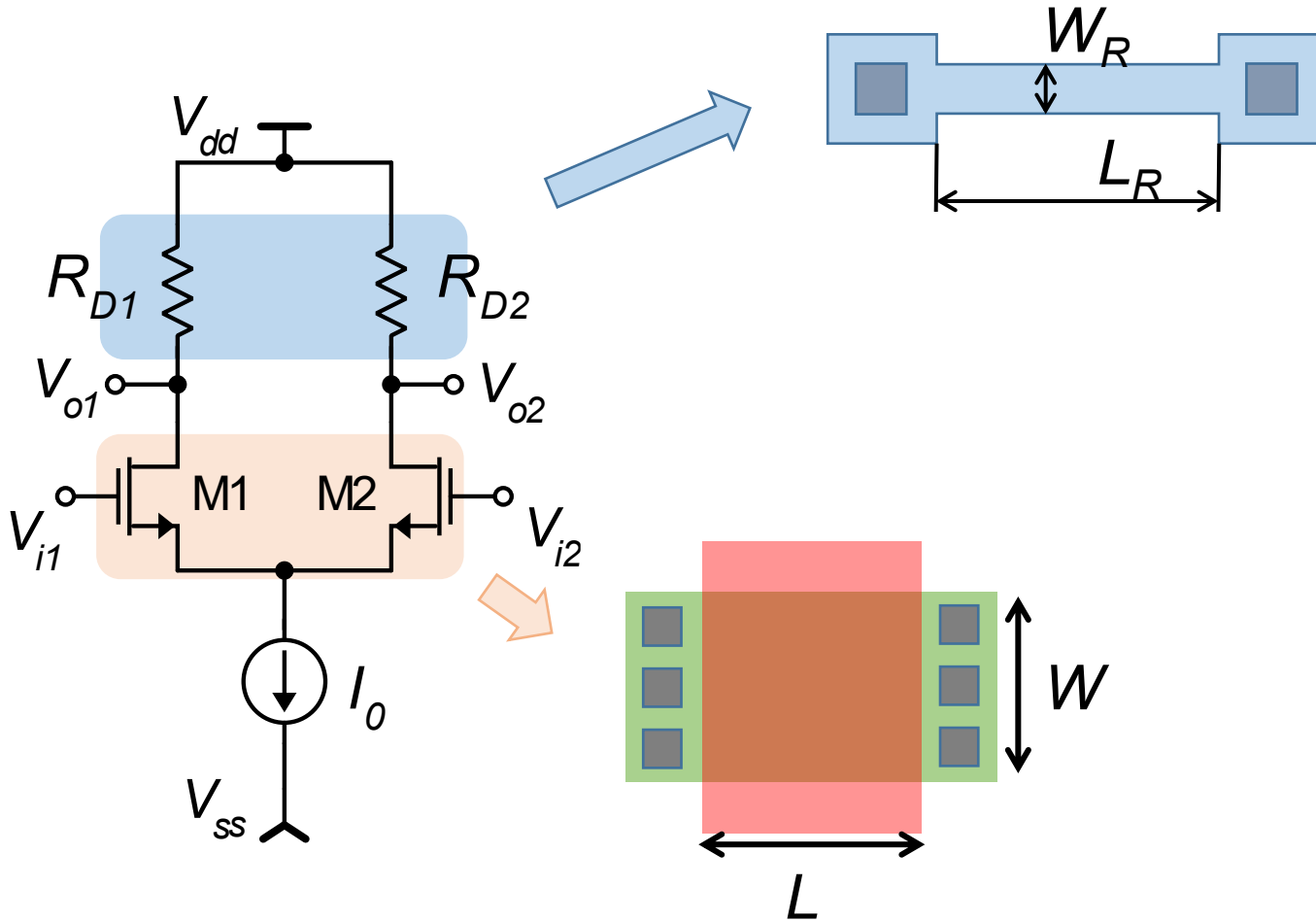


$$V_{out} = V_{OD} = \underbrace{R_{D2}I_{D2} - R_{D1}I_{D1}}_{\Delta Z = V_{out}}$$

defining: $Z = R_D I_D$

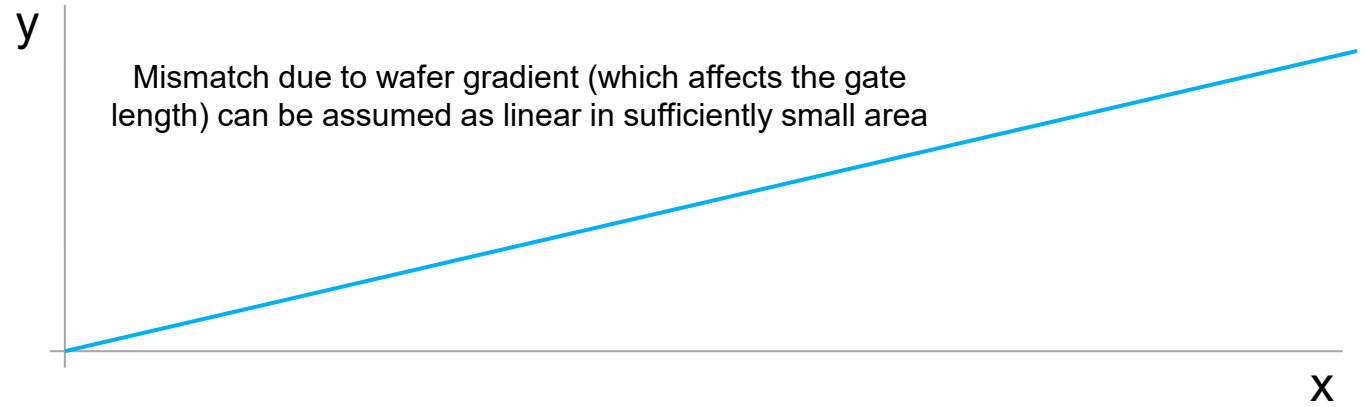
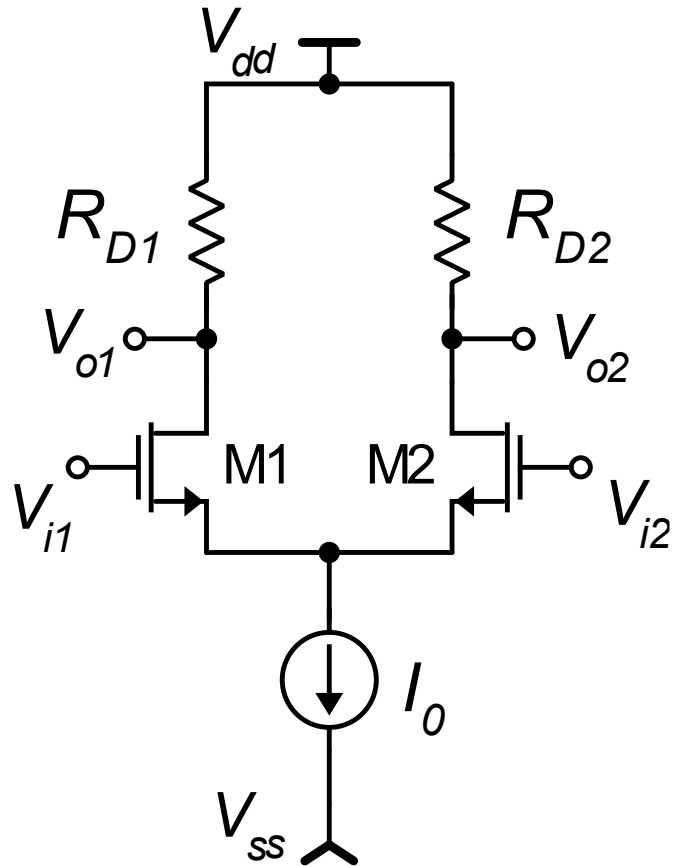
Imposing $V_{out}=0$ is the same as imposing $\Delta Z=0$

Role of the design parameters

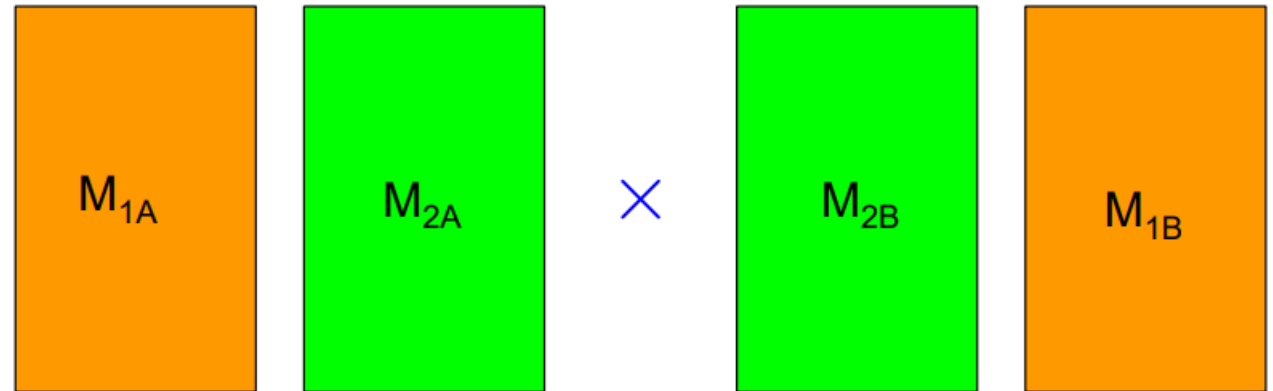


Small offset means large silicon area

How to improve mismatch



Layout of M1 and M2



Same methodology can be applied to the load resistors R_{D1} and R_{D2} !

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.